DESCRIPTION OF COURSEWORK

Course Code	EEE326*
Course Name	Laboratory on Electrical Power Engineering
Lecturer	Atifa Faaidha Binti Munauwer
Academic Session	2023/09
Assessment Title	Final Project (Individual)

A. Introduction/ Situation/ Background Information

The electrical power system encompasses a comprehensive network designed for the generation, transmission, and distribution of electrical energy. Within this system, transmission lines serve as vital conduits, facilitating the high-voltage transportation of electricity over long distances from power plants to load centers. As part of the power system, the IEEE 5-bus system stands as a benchmark model, featuring five buses interconnected by transmission lines. It serves as a simple yet representative model for testing various power system analysis and optimization techniques. The 5-bus system is often used to study power flow, economic dispatch, contingency analysis, and other aspects of power system operation. This system is often represented in the form of a one-line diagram, showing the buses and transmission lines connecting them. Power flow analysis can be performed on this system to determine the steady-state operating conditions, and various studies can be conducted to analyse the system's behaviour under different scenarios.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 1 Perform power engineering related experiments by using relevant equipment and tools with consideration for safety

CLO 2 Demonstrate the ability to communicate effectively the findings and outcomes of laboratory activities in written reports

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

1. This is an individual final project.
2. This project will contribute to 50% of your total assessment for this course
3. You are required to use Transmission line trainer in A1#214 laboratory, Powerworld software and MATLAB Simulink to complete this final project.
4. You must prepare one softcopy of a typewritten report detailing the answers to the task. Save the report as EEE2004123_NAME.pdf. Upload the softcopy of your report, simulation file, and Turnitin report into submission folder created for you in the Moodle system.
5. Submission deadline: 29 December 2023 (Week 15), 11:59:59 pm

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Practical (Part A) [CLO1-PLO5:P3]	30
2.	Powerworld Simulation (Part B) [CLO1-PLO5:P3]	15
3.	MATLAB Simulink Simulation (Part B) [CLO1-PLO5:P3]	15
4.	Project Report (Part 3)[CLO2-PLO10:A3]	40
	TOTAL	**100**

F. Task(s)

This final project consists of three (3) part. Answer ALL part.

Part A: [CLO1-PLO5:P3] [30 Marks]

In power systems, the concept of a matched load plays a pivotal role in optimizing the efficiency of power transfer from a source to a load. Achieving a matched load involves aligning the impedance of the load with that of the source, thereby enabling the system to transfer power with minimal losses.

a. By using the GOTT Transmission Line Trainer, perform a matched load technique to determine the value of resistive load at which the line no longer consumes any reactive power. The line resistance is 13Ω per phase and its inductance is 290mH per phase. [CLO1-PLO5:P3] [8 Marks]

b. By using resistive load from (a), construct a circuit to determine the voltage profiles, line flows, efficiency and power losses within the system. [CLO1-PLO5:P3] [10 Marks]

c. Build the system model using MATLAB Simulink software and compare with practical results. [CLO1-PLO5:P3] [12 Marks]

Part B:[CLO1-PLO5:P3][30 Marks]

Power flow analysis is a fundamental analysis technique used in power system studies to calculate the steady-state voltages, line currents, and power flows in a network. As a power engineer, you are required to perform power flow analysis of the IEEE 5-bus system using Powerworld Simulator and MATLAB Simulink.

The task specification of the project is given below:

1. **Model Creation:**
   a. Perform a power flow analysis of the IEEE 5-bus system shown in Figure 1 using Powerworld Simulator to determine the voltage profiles, line flows, and power losses within the system. The system parameter is given in Table 1 and Table 2.
   [CLO1-PLO5:P3] [15 Marks]

   b. Build the IEEE 5-bus system model using MATLAB Simulink software and compare with Powerworld simulation results. [CLO1-PLO5:P3] [15 Marks]

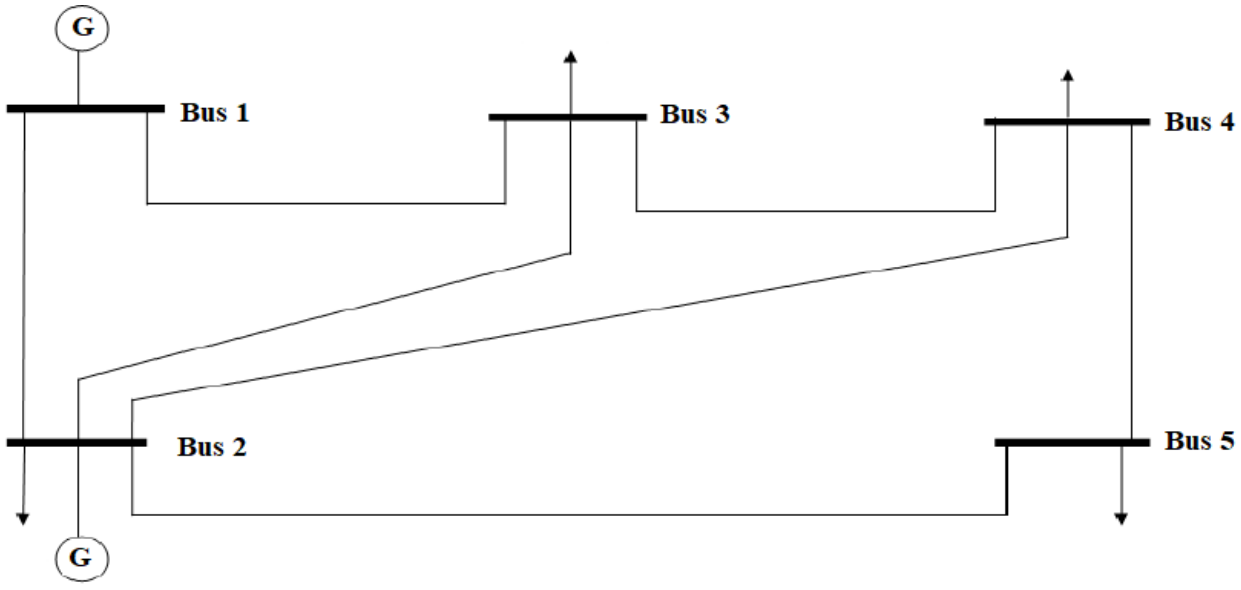


Figure 1: IEEE 5 Bus System

Bus No	Bus Type	Voltage (p.u)	Generation		Load	
			MW	Mvar	MW	Mvar
1	Swing	1.06	0	0	0	0
2	PV	1.00	40	30	20	10
3	PQ	1.00	0	0	45	15
4	PQ	1.00	0	0	40	5
5	PQ	1.00	0	0	60	10
*Sbase = 100MVA						

Table 1 Bus Data of IEEE 5 Bus System

Line From	Line To	R (p.u)	X(p.u)	B(p.u)
1	2	0.02	0.06	0.03
1	3	0.08	0.24	0.025
2	3	0.06	0.18	0.02
2	4	0.06	0.18	0.02
2	5	0.04	0.12	0.015
3	4	0.01	0.03	0.01
4	5	0.08	0.24	0.025

Table 2 Line or Branch Data of IEEE 5 Bus System

Part C: Final Report [CLO2-PLO10:A3] [40 Marks]

Report your findings according to the following:

i. Report your finding for both Part A and Part B in **ONE** technical report. Display all simulation snapshot results in your report. Present the power flow analysis results, including bus voltage profiles, line flows, power losses, and any additional relevant information obtained from the GOTT Trainer, Powerworld Simulator and MATLAB Simulink.

ii. Explain the power flow analysis results to gain insights into the system behavior and performance.

iii. Study and explain the power flow analysis findings, identifying any voltage deviations, heavily loaded lines, reactive power requirements, or other important system characteristics.

iv. Propose any potential system improvements or corrective actions based on the analysis results.

v. Report Guidelines:

Font type: Times New Roman, Font size: 12pt.

Line Spacing: 1.5

Layout: Single-column page layout.

Suggested topics:

- Introduction
- Project background
- Objectives
- Methodology
- Results and discussion
- Learning outcome achievement
- Conclusions and recommendations

Submission

You are required to submit:

i. Project report in PDF format. Save the report as EEE2004123_NAME.pdf

ii. PowerWorld Simulator Model: Provide the complete PowerWorld Simulator model files (.pwb and .pwd) representing your simulations, including all defined components and their settings.

iii. MATLAB Simulink Model: Provide the complete Simulink Model (.slx) representing your simulation system, including all defined components and their settings.

iv. Include your plagiarism declaration and turnitin report together with your lab report. Note that plagiarism is strictly prohibited: your turnitin report must be below 10%.

APPENDIX 1

MARKING RUBRICS

Component Title	Final Report [CLO2-PLO10:A3][40 Marks]				**Percentage (%)**	40	
Criteria	**Score and Descriptors**					**Weight (%)**	**Marks**
	Excellent (5)	**Good (4)**	**Satisfactory (3)**	**Inadequate (2)**	**Poor (1)**		
Introduction & Background	Well-written and concise background of the study with clear definition of key terms and engineering concepts	A good background that includes some definitions of key terms and engineering concepts	The background study is relevant but does not define key terms and engineering concepts.	The background study is partially relevant.	The background study is irrelevant.	2	
Objectives	The objective are clearly identified and well stated	The objectives are clear	Somewhat well-defined objectives	Poorly defined objectives.	The objectives are not relevant	2	
Methodology	All experimental details are covered.	Important experimental details are covered, some minor details missing	Missing some important experimental details	Missing several important experimental details	All-important experimental details are missing	2	
Presentation of findings/results	Clear and comprehensive presentation of results/findings.	Results/findings were presented but incomplete	Results/findings were presented but inadequate.	Lack of presentation of results/findings.	No presentation of findings/results.	4	
Discussion	Details and outputs are justified and labelled clearly. The findings are discussed excellently concerning theory with sufficient evidence.	Details outputs explained and labelled sufficiently. The findings are discussed concerning theory and demonstrate appropriate evidence	Details and outputs are fairly explained and labelled. The findings are fairly discussed concerning theory and show partially appropriate evidence	Details and outputs are poorly explained and labelled. The findings are poorly discussed concerning theory and show partially but incomplete evidence	Details and outputs are not explained and labelled clearly. The findings are not discussed concerning theory and inappropriate evidence.	5	
Conclusion and Recommendation	Conclusion covers all the important points and directly relates the findings to the objectives	Conclusion covers most of the important points and adequate relation of the findings to the objectives	Conclusion on major points are drawn, but less value of the relation of the findings to the objectives	Inappropriate conclusion with little or no relation of the finding to the objectives.	Failed to write the conclusion.	3	
Organization, language, and formatting	Very well-organized report. Excellent written grammar without error The format is consistent, figures and tables are neatly done	Well-organized report. Well-written grammar. The format is generally consistent, figures and tables are neatly done.	The report is organized Appropriate written grammar. Mostly consistent format of figures and tables with some spelling mistakes.	Inconsistent format such as font type, size, and spacing. Figures and tables are disorganized and several grammar mistakes	Unorganized report Poor use of grammar and language.	2	
					Total (100 Marks)		/100
					Total (40%)		

Note to students: Please print out and attach this appendix together with the submission of coursework